

Title:

SYSTEM AND METHOD FOR AUTOMATED EDUCATIONAL CONTENT GENERATION AND STUDENT EVALUATION

Abstract:

A system and method for automated educational content generation and student evaluation are disclosed. The system includes a content generation module that dynamically creates educational content tailored to a specified subject, grade level, and topic. A quiz creation module automatically produces quizzes instantly, aligned with corresponding lessons, including multiple choice options and correct answers. A student evaluation module scores responses, generates detailed feedback reports, analyzes student mistakes, and provides personalized study recommendations based on the analysis. The integrated approach streamlines content delivery, assessment, and individualized learning pathways, often leveraging natural language processing models across the modules.

Field of the Invention:

The present invention relates generally to educational technology, and more particularly, to systems and methods for automated generation of educational content, automated creation of quizzes, and automated evaluation of student performance with personalized feedback and study recommendations.

Background of the Invention:

In traditional educational settings, the creation of teaching materials, lessons, and assessments is a labor-intensive and time-consuming process. Educators often spend significant time developing content tailored to specific subjects, grade levels, and topics, as well as designing quizzes that accurately reflect the learned material. Furthermore, the evaluation of student responses, especially for a large number of students, can be a daunting task, making it challenging to provide timely, detailed, and personalized feedback.

Existing e-learning platforms and educational software tools often offer static content or pre-defined

assessments, which may not adequately address the diverse learning needs and knowledge gaps of individual students. While some systems provide basic scoring, they frequently lack the capability for in-depth analysis of student mistakes or the provision of truly personalized study recommendations beyond generic suggestions. The absence of dynamic content generation and adaptive assessment tools limits the efficiency and effectiveness of personalized learning experiences. There is a need for an integrated system that can automate the entire cycle from content creation to personalized student evaluation, thereby enhancing educational outcomes and reducing the burden on educators.

Summary of the Invention:

The present invention provides a novel system and method for automated educational content generation and student evaluation. The system comprises a content generation module configured to dynamically generate educational content that is automatically tailored based on a specified subject, a specified grade level, and a specified topic. This module can synthesize information from a plurality of educational materials to create comprehensive and relevant lessons.

The system further includes a quiz creation module configured to automatically create quizzes that are instantly produced and aligned with their corresponding lessons. These quizzes typically include multiple choice options and designated correct answers, with alignment achieved by classifying textual content of the lesson and the quiz based on subject, grade level, and topic.

Additionally, the system incorporates a student evaluation module designed to score student responses, generate detailed feedback reports including correct answers, perform in-depth analysis of student mistakes (including processing responses with variations or errors), and provide personalized study recommendations based on this analysis. These recommendations may include suggested alternative learning resources or targeted practice exercises. Each of these modules, or a combination thereof, may utilize a natural language processing model to facilitate their respective functions.

The method aspects of the invention mirror the system's capabilities, encompassing steps for

dynamically generating tailored educational content, automatically creating aligned and instant quizzes, and comprehensively evaluating student responses through scoring, detailed feedback, mistake analysis, and personalized recommendation provision. The invention aims to streamline the educational process, enhance personalization, and improve learning outcomes.

Detailed Description:

The present invention relates to a comprehensive system and method for automating various aspects of educational content creation and student evaluation. The system integrates several modules designed to work synergistically to provide a dynamic and personalized learning experience.

In one embodiment, the system comprises a content generation module, a quiz creation module, and a student evaluation module. These modules are typically implemented as software components running on one or more computing devices, which may include servers, cloud-based platforms, personal computers, or mobile devices.

The content generation module is configured to dynamically generate educational content. This generation process is automated and intelligently tailored based on specific input parameters: a specified subject (e.g., mathematics, history, biology), a specified grade level (e.g., 5th grade, high school, university freshman), and a specified topic within that subject (e.g., "fractions" within mathematics, "World War II" within history). The module receives these parameters as input and then proceeds to synthesize relevant information. This synthesis can involve drawing from a plurality of educational materials, such as textbooks, academic papers, online articles, and curated databases. The module employs advanced algorithms, potentially including natural language processing (NLP) models, to understand, extract, summarize, and reformulate information into coherent and pedagogically sound educational content. The dynamic nature ensures that new content can be generated on demand for virtually any combination of subject, grade level, and topic, providing flexibility and up-to-date information.

The quiz creation module operates in conjunction with the content generation module or based on existing lessons. It is configured to automatically create a quiz that is instantly produced. This immediate generation capability allows for on-the-fly assessment creation aligned with a corresponding lesson. The quizzes are designed to include multiple choice options for each question and clearly designated correct answers. To ensure alignment, the quiz creation module employs techniques such as textual content classification. This involves analyzing the textual content of both the lesson and the generated quiz questions to ensure they share a common subject, grade level, and topic. For instance, an NLP model might be used to extract keywords, concepts, and semantic relationships from the lesson content, and then generate quiz questions that directly test understanding of those elements.

The student evaluation module is a critical component for assessing learning and providing support. This module is configured to perform several key functions:

First, it scores student responses to the quizzes. This scoring can be automated by comparing student-selected answers against the designated correct answers for multiple-choice questions.

Second, it generates detailed feedback reports. These reports are comprehensive and include not only the student's score but also the correct answers for each question, allowing students to understand where they went wrong.

Third, the module performs analysis of student mistakes. This analysis goes beyond simply identifying incorrect answers. It processes student responses that may include variations or errors, attempting to understand the nature of the mistake rather than just marking it as wrong. For example, if a student provides a slightly misphrased but semantically correct answer in an open-ended scenario (if supported), the module might recognize the underlying intent. This analysis helps in pinpointing common misconceptions or areas of weakness.

Fourth, based on the analysis of student mistakes, the module provides personalized study recommendations. These recommendations are tailored to the individual student's specific learning needs and identified areas of difficulty. For instance, if a student consistently struggles with a

particular concept, the recommendations might include suggested alternative learning resources (e.g., different explanations, videos, articles) or targeted practice exercises focusing specifically on that concept. This personalized approach aims to guide students efficiently towards mastery.

In various embodiments, the content generation module, the quiz creation module, and the student evaluation module each utilize a natural language processing (NLP) model. These NLP models can be large language models (LLMs), transformer networks, or other machine learning models trained on vast datasets of text and educational content. For the content generation module, NLP models enable the understanding, summarization, and generation of human-like text for lessons. For the quiz creation module, NLP models facilitate question generation, distractor selection, and alignment verification. For the student evaluation module, NLP models assist in understanding student responses, analyzing semantic errors, and generating contextually relevant feedback and recommendations.

A method for automated educational content generation and student evaluation is also provided. This method comprises the steps of:

1. Dynamically generating educational content, wherein the educational content is automatically tailored based on a specified subject, a specified grade level, and a specified topic.
2. Automatically creating a quiz, wherein the quiz is instantly produced, aligned with a corresponding lesson, and includes multiple choice options and designated correct answers.
3. Evaluating student responses by:
 - a. Scoring the student responses.
 - b. Generating detailed feedback reports including correct answers.
 - c. Performing analysis of student mistakes.
 - d. Providing personalized study recommendations based on the analysis of student mistakes.

The method can further include steps corresponding to the detailed functionalities of the system modules, such as synthesizing information from a plurality of educational materials for content

generation, classifying textual content for quiz alignment, processing student responses with variations or errors for mistake analysis, and suggesting alternative learning resources or targeted practice exercises for study recommendations. The method leverages computational power and artificial intelligence to create an efficient, adaptive, and highly personalized learning environment.

Claims:

1. A system for automated educational content generation and student evaluation, comprising:
 - * a content generation module configured to dynamically generate educational content, wherein the educational content is automatically tailored based on a specified subject, a specified grade level, and a specified topic;
 - * a quiz creation module configured to automatically create a quiz, wherein the quiz is instantly produced, aligned with a corresponding lesson, and includes multiple choice options and designated correct answers; and
 - * a student evaluation module configured to:
 - * score student responses;
 - * generate detailed feedback reports including correct answers;
 - * perform analysis of student mistakes; and
 - * provide personalized study recommendations based on the analysis of student mistakes.
2. The system of claim 1, wherein the content generation module is further configured to synthesize information from a plurality of educational materials to generate the educational content.
3. The system of claim 1, wherein the quiz creation module is configured to align the quiz with the corresponding lesson by classifying textual content of the lesson and the quiz based on subject, grade level, and topic.
4. The system of claim 1, wherein the student evaluation module is configured to perform analysis of student mistakes by processing student responses that include variations or errors.
5. The system of claim 1, wherein the personalized study recommendations include suggested

alternative learning resources or targeted practice exercises.

6. The system of claim 1, wherein the content generation module, the quiz creation module, and the student evaluation module each utilize a natural language processing model.

7. A method for automated educational content generation and student evaluation, the method comprising:

- * dynamically generating educational content, wherein the educational content is automatically tailored based on a specified subject, a specified grade level, and a specified topic;
- * automatically creating a quiz, wherein the quiz is instantly produced, aligned with a corresponding lesson, and includes multiple choice options and designated correct answers; and
- * evaluating student responses by:
 - * scoring the student responses;
 - * generating detailed feedback reports including correct answers;
 - * performing analysis of student mistakes; and
 - * providing personalized study recommendations based on the analysis of student mistakes.